CHAPTER 5 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

Intelligence Gathering, Red Teaming, and Adversarial Assessment of UAS

Companion figures, evidence interpretation, detection metrics, and a collaborative assessment exercise

Learning goal: produce defensible findings from a scoped and repeatable security assessment.

UAS security assessment lifecycle

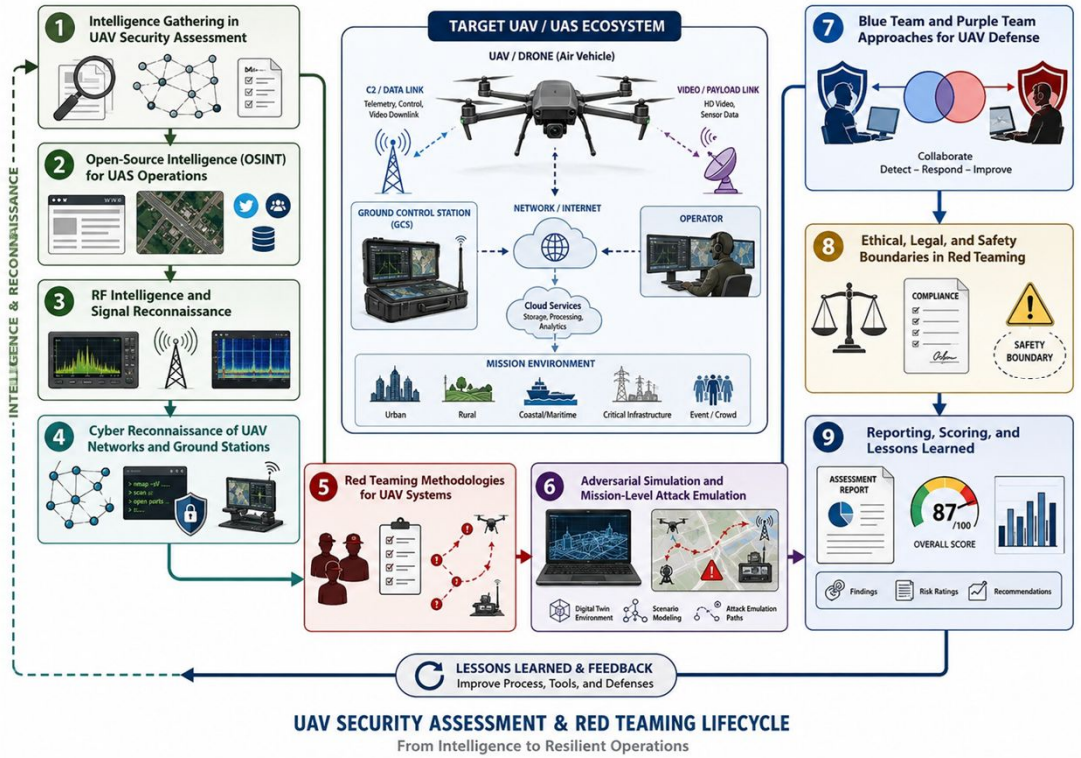


Figure S5.1. Planning, collection, analysis, controlled testing, and reporting form an iterative assessment. Findings should connect observations to mission consequences and corrective action.

Source illustration is conceptual. Teaching notes identify assumptions, terminology, and evidence limits.

RF observations and intelligence products

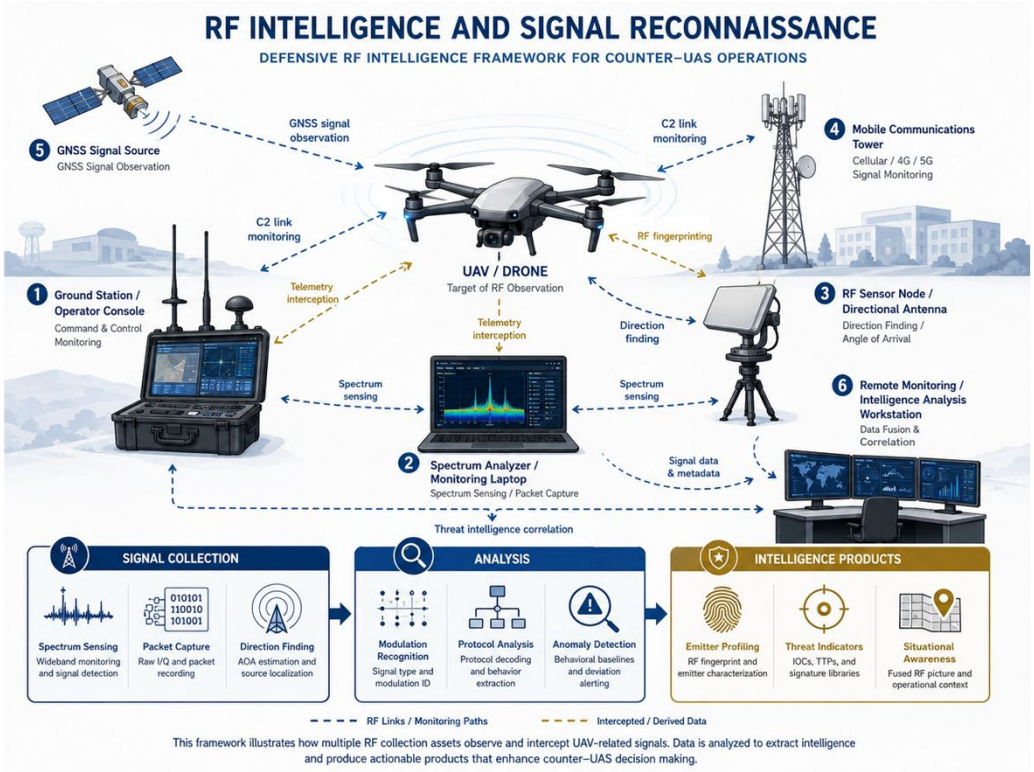


Figure S5.2. Signal observations can support hypotheses about emitters and activity. Identification, aircraft association, and attribution require corroborating evidence.

Source illustration is conceptual. Teaching notes identify assumptions, terminology, and evidence limits.

Network observations and operating context

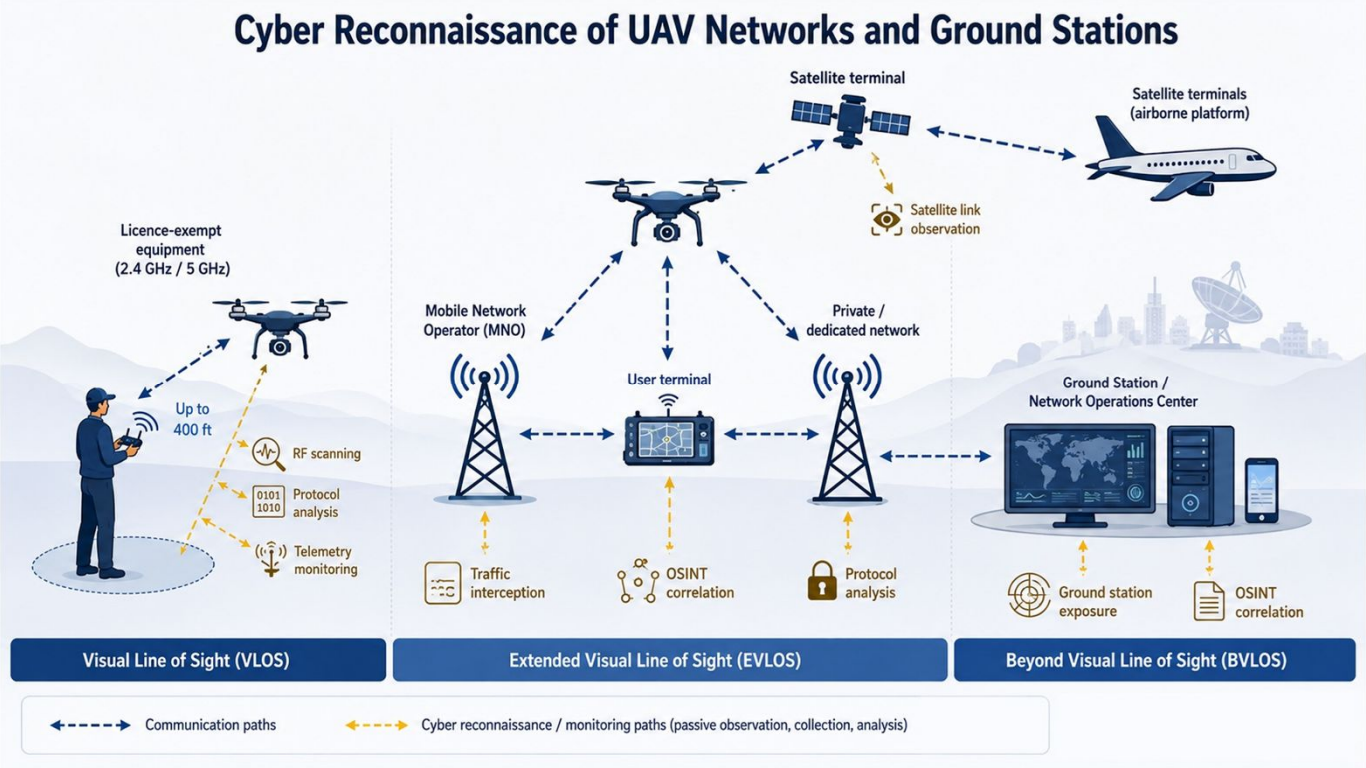


Figure S5.3. Network topology and monitoring access determine the available evidence. VLOS/BVLOS describe visual-contact conditions independently of the transport technology.

Source illustration is conceptual. Teaching notes identify assumptions, terminology, and evidence limits.

Blue-team defense and purple-team collaboration

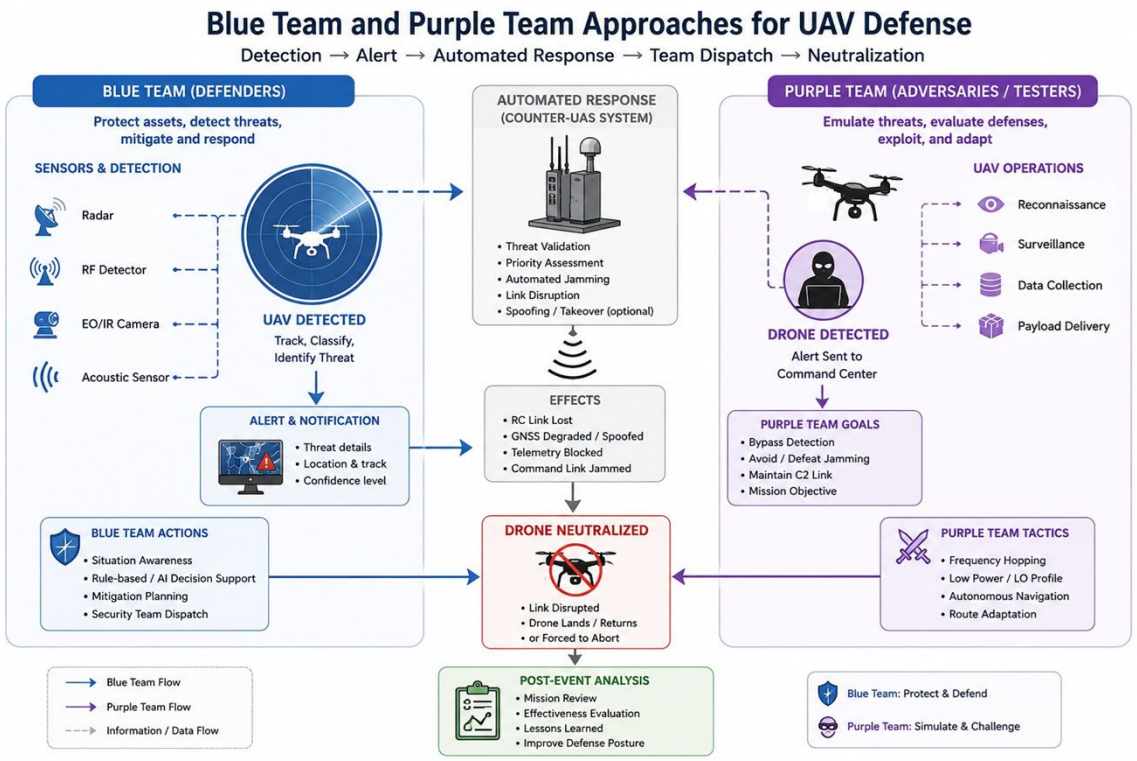


Figure S5.4. Purple teaming combines testers and defenders to validate detection and response. Keep threat detection, authorization, and operational intervention as separate decisions.

Source illustration is conceptual. Teaching notes identify assumptions, terminology, and evidence limits.

Data-link exposure and assessment hypotheses

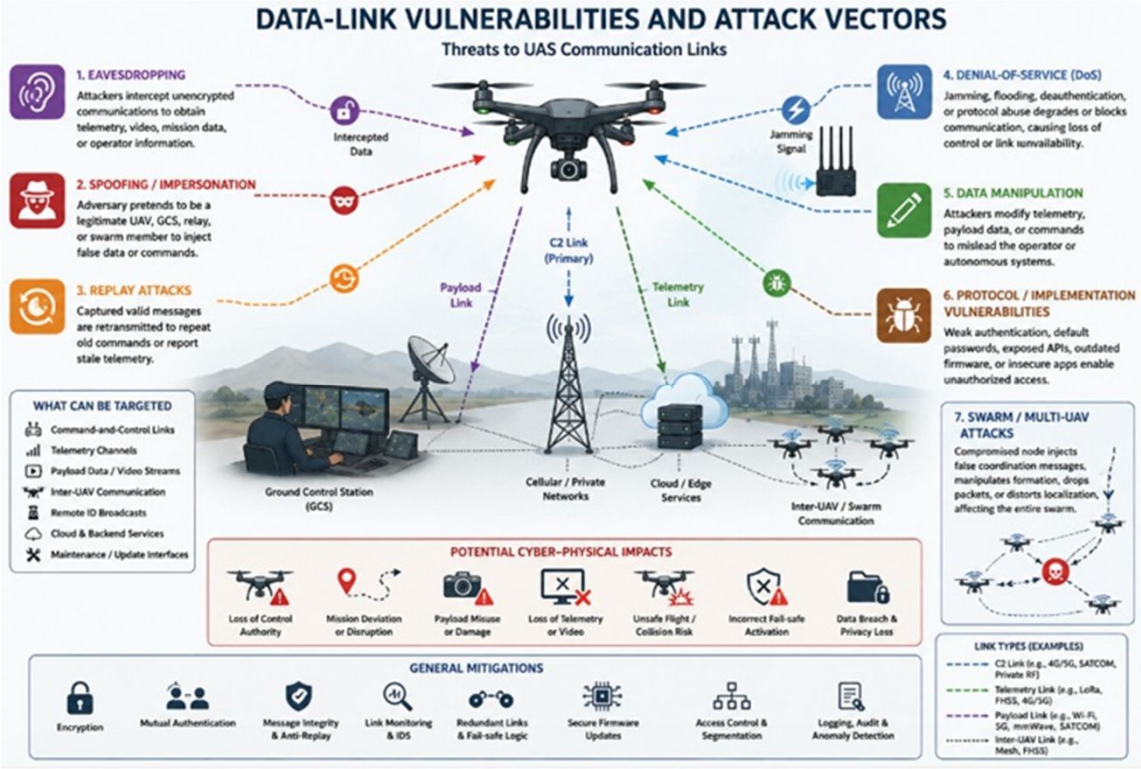


Figure S5.5. Candidate threat paths guide test selection. Each finding needs evidence of the relevant preconditions, affected interface, and observed mission effect.

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Assessment scope and team responsibilities

A test plan assigns ownership and defines which actions and evidence belong to the assessment.

Role / responsibility	Contribution	Record to retain
Assessment owner	Defines objectives, scope, contacts, exclusions, and stop conditions	Approved test plan and change / restoration record
Red team / assessors	Execute scoped test scenarios and document actual test actions	Scenario IDs, timestamps, preconditions, and observed effects
Blue team / defenders	Monitor, investigate, and apply the agreed response process	Alerts, investigation records, response actions, and limitations
Purple-team collaboration	Jointly replay evidence and refine detections, controls, and procedures	Evidence comparison, gaps, corrective actions, and retest results

Success means a useful, reproducible finding and a validated improvement, rather than a score alone.

Observations, inferences, and decisions

Report what the evidence supports and state the uncertainty that affects interpretation.

Evidence / claim	Supported interpretation	Additional evidence needed
Unusual RF energy	A change in the monitored spectrum or received power	Calibration, baseline, timing, and alternative-source checks
Repeated protocol pattern	Traffic has observable structure consistent with a candidate class	Validated classifier performance and independent aircraft association
Link outage or track deviation	A communication or navigation performance anomaly occurred	Synchronized system logs and a defensible causal explanation
Claim of hostile intent or operator identity	A hypothesis requiring evidence beyond the signal anomaly	Corroborated attribution, context, and the appropriate review process

A defensible assessment preserves competing explanations and records confidence explicitly.

Worked example: detection performance

Synthetic dataset: 100 fixed observation windows, including 20 simulated threat events and 80 benign windows.

Ground-truth label	Alert	No alert
Simulated threat event	16 true positives (TP)	4 false negatives (FN)
Benign window	4 false positives (FP)	76 true negatives (TN)

Precision = TP / (TP + FP) = 16 / 20 = 80%

Recall = TP / (TP + FN) = 16 / 20 = 80%

False-positive rate = FP / (FP + TN) = 4 / 80 = 5%

These values describe the synthetic test set. Operational performance depends on event prevalence and conditions.

Practical exercise: collaborative detection validation

Use a recorded dataset or a controlled simulator with known event labels and a repeatable baseline.

1. Scope and baseline	Choose one observable failure or threat hypothesis. Define the data, controls, and stop conditions.
2. Controlled scenarios	Compare benign variation with a simulated anomaly such as a stale update or a link outage.
3. Joint evidence review	Match scenario times to alerts, system state, and mission effects. Identify misses and false alarms.
4. Improvement and retest	Change one detection or control, repeat the scenarios, and document benefits and remaining gaps.

Deliverable: a scoped test plan, labeled event log, detection metrics, and a retest report.

References and terminology

Primary references support assessment planning, adversary emulation, and response analysis.

NIST SP 800-115 Technical Guide to Information Security Testing and Assessment (2008). csrc.nist.gov/pubs/sp/800/115/final

MITRE ATT&CK Adversary Emulation Plans. Behavior-based assessment examples. attack.mitre.org/resources/adversary-emulation-plans/

NIST SP 800-61r3 Incident Response Recommendations and Considerations for Cybersecurity Risk Management (2025). csrc.nist.gov/pubs/sp/800/61/r3/final

EASA Easy Access Rules for Unmanned Aircraft Systems, June 2026. Operational terminology and visual-contact definitions.

OSINT: open-source intelligence RF: radio frequency UAS: unmanned aircraft system
TP / FP: true / false positives FN / TN: false / true negatives